Titanic EDA Report

Exploratory Data Analysis of the Titanic Passenger Dataset

Dataset: 891 passengers, 12 features

Survival Rate: 38.4%

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1. Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) of the Titanic passenger dataset. The dataset contains information on 891 passengers aboard the RMS Titanic, including demographics, ticket class, fare paid, and survival outcome. The analysis aims to identify patterns, correlations, and factors that influenced passenger survival during the disaster.

Key Statistics

891
Total Passengers

38.4%
Survival Rate

29.7
Average Age

$32.20
Average Fare

2. Dataset Overview

The Titanic dataset consists of 891 records with 12 features. Below is a summary of each column including data types and basic statistics.

Column Descriptions

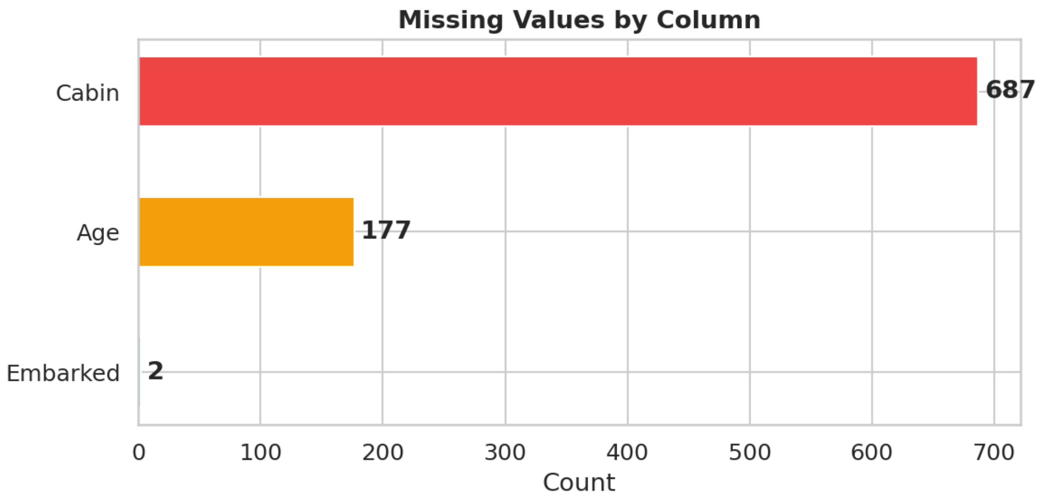
Column	Type	Description
PassengerId	int	Unique identifier
Survived	int	0 = No, 1 = Yes
Pclass	int	Ticket class (1/2/3)
Name	str	Passenger name
Sex	str	Gender
Age	float	Age in years
SibSp	int	Siblings/spouses aboard
Parch	int	Parents/children aboard
Ticket	str	Ticket number
Fare	float	Passenger fare
Cabin	str	Cabin number
Embarked	str	Port of embarkation

Summary Statistics

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Age	714.0	29.7	14.53	0.42	20.12	28.0	38.0	80.0
Fare	891.0	32.2	49.69	0.0	7.91	14.45	31.0	512.33
SibSp	891.0	0.52	1.1	0.0	0.0	0.0	1.0	8.0
Parch	891.0	0.38	0.81	0.0	0.0	0.0	0.0	6.0

3. Missing Value Analysis

Understanding missing data is critical before any analysis. Three columns contain missing values, with Cabin being the most severely affected at 77.1% missing.



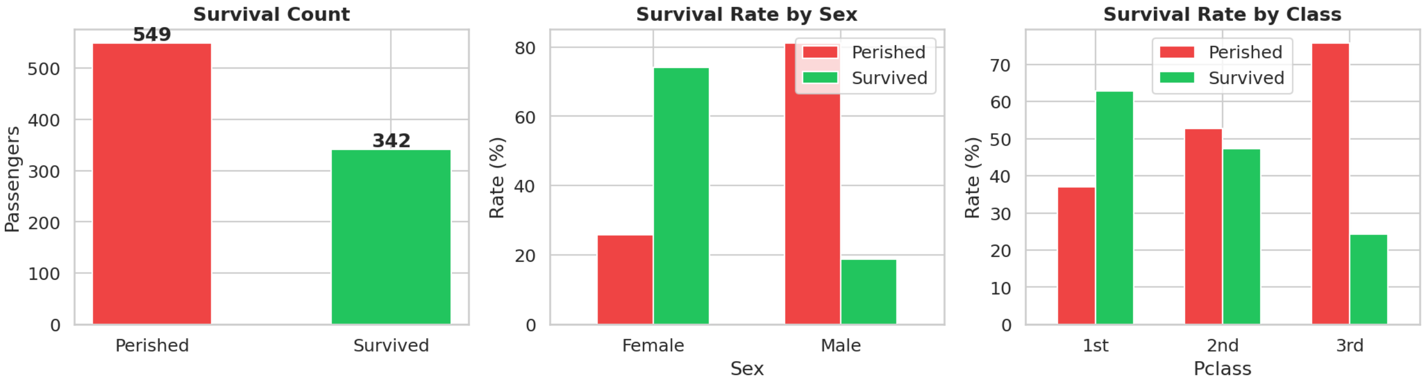
Missing Value Details

Column	Missing	Percent	Severity
Cabin	687	77.1%	High
Age	177	19.9%	Medium
Embarked	2	0.2%	Low

Recommendation: Cabin should be dropped or heavily engineered (e.g., extract deck letter). Age can be imputed using median values stratified by class and gender. The 2 missing Embarked values can be filled with the mode (Southampton).

4. Survival Analysis

Overall, only 38.4% of passengers survived the disaster. Survival was heavily influenced by gender and ticket class, reflecting the 'women and children first' protocol and the socioeconomic hierarchy aboard the ship.



Survival by Sex

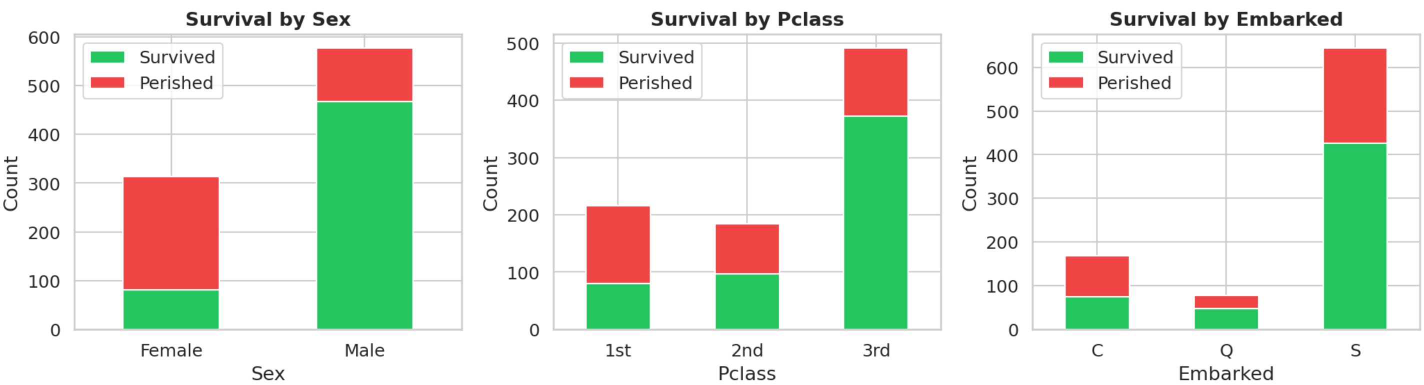
Sex	Passengers	Survival Rate (%)
female	314	74.2%
male	577	18.9%

Women had a dramatically higher survival rate (74.2%) compared to men (18.9%). This is the strongest single predictor of survival in the dataset.

Survival by Class

Class	Passengers	Survival Rate (%)
1st Class	216	63.0%
2st Class	184	47.3%
3st Class	491	24.2%

First-class passengers survived at 63.0%, compared to only 24.2% for third-class. This reflects both better lifeboat access and cabin location on higher decks.



5. Demographic Analysis

Passenger Distribution by Class

- Class 1: 216 passengers (24.2%)
- Class 2: 184 passengers (20.7%)
- Class 3: 491 passengers (55.1%)

Passenger Distribution by Embarked Port

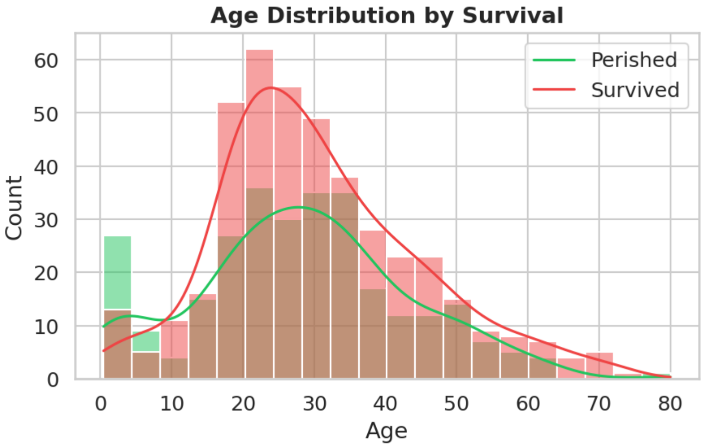
- Southampton: 644 passengers (72.3%)
- Cherbourg: 168 passengers (18.9%)
- Queenstown: 77 passengers (8.6%)

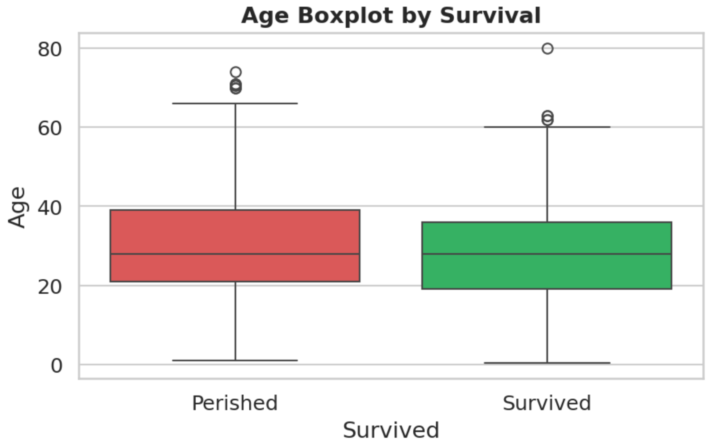
Gender Distribution

- Male: 577 passengers (64.8%)
- Female: 314 passengers (35.2%)

6. Age Analysis

Age is an important factor in survival, with children being given priority during evacuation. The average age of passengers was 29.7 years, with 177 missing values (19.9%).

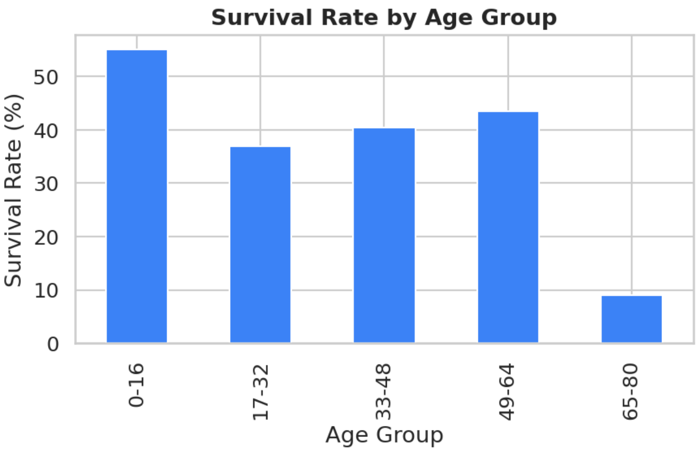


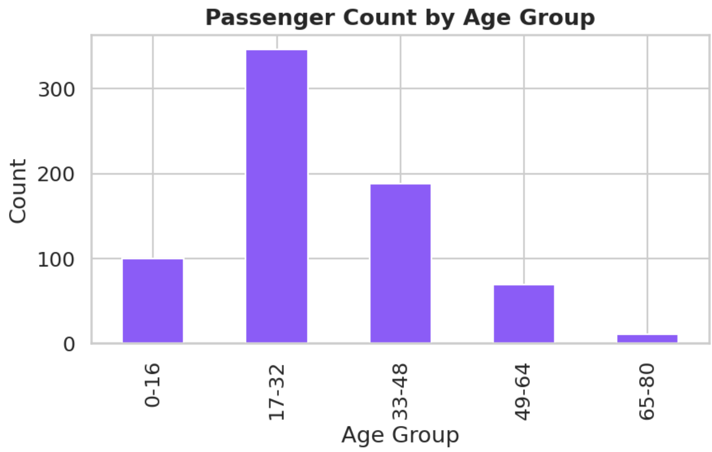


Age Statistics

- Mean age: 29.7 years
- Median age: 28.0 years
- Age range: 0.4 to 80.0 years
- Standard deviation: 14.5 years

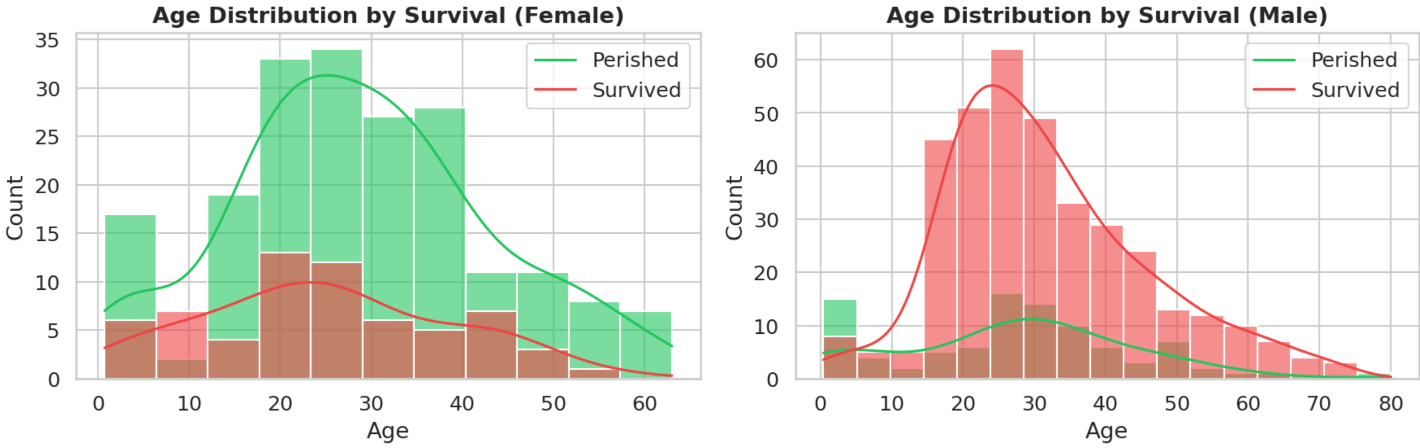
Survival by Age Group





Age Range	Passengers	Survival Rate (%)
(0.0, 16.0]	100	55.0%
(16.0, 32.0]	346	37.0%
(32.0, 48.0]	188	40.4%
(48.0, 64.0]	69	43.5%
(64.0, 80.0]	11	9.1%

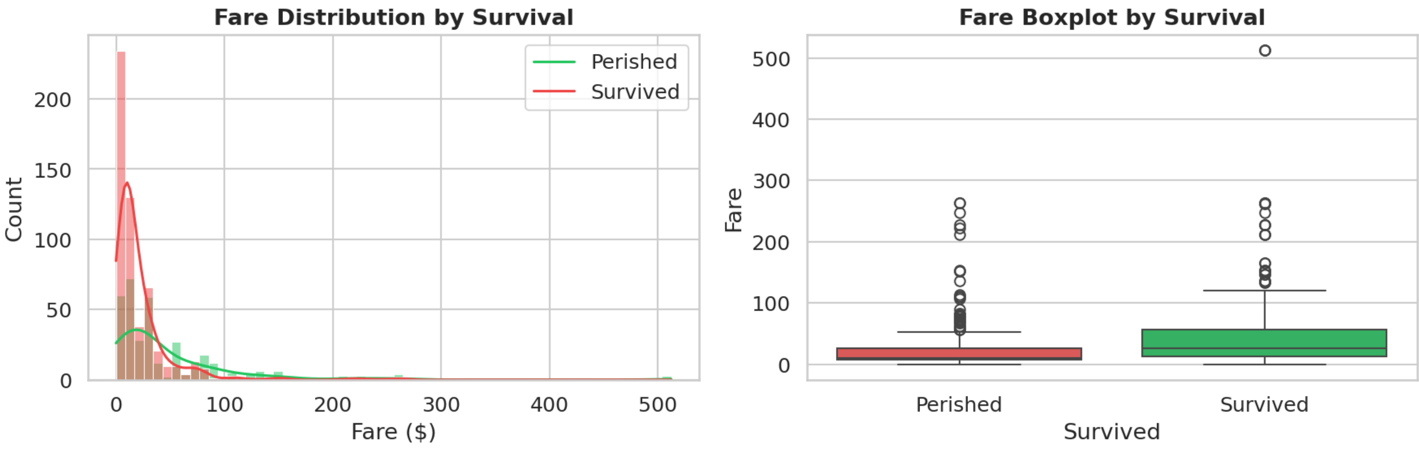
Children aged 0-16 had the highest survival rate at 55.0%, confirming the 'children first' evacuation policy. The oldest age group (65-80) had the lowest survival rate at 9.1%.



Separating by gender reveals that young female passengers had the highest survival rates, while male children also fared better than adult males.

7. Fare Analysis

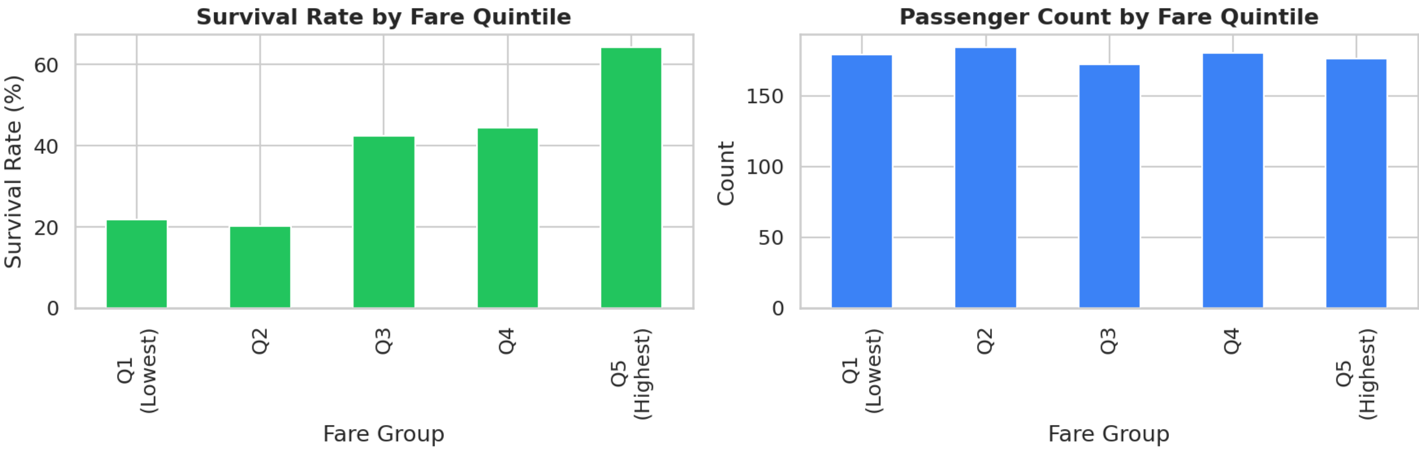
Fare paid is strongly correlated with survival (r=0.257), as higher fares indicate higher class tickets with better cabin locations and lifeboat access.



Fare Statistics

- Mean fare: $32.20
- Median fare: $14.45
- Fare range: $0.00 to $512.33
- Standard deviation: $49.69

Survival by Fare Quintile

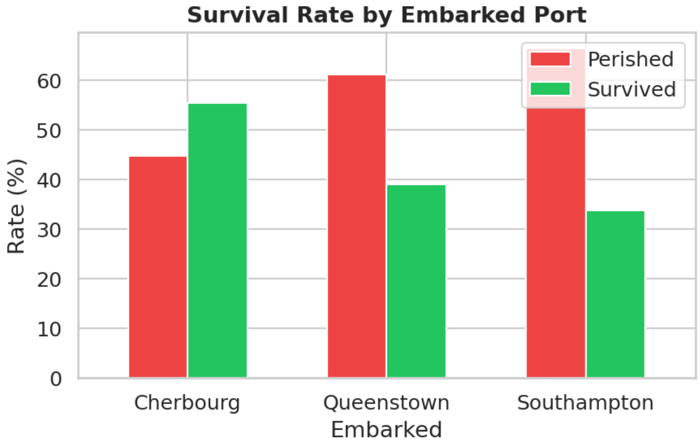


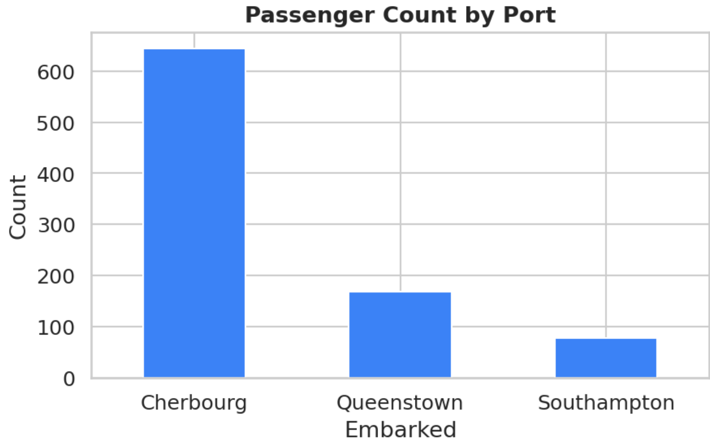
Fare Range	Passengers	Survival Rate (%)
(-1.0, 102.0]	838	36.2%
(102.0, 205.0]	33	75.8%
(205.0, 307.0]	17	64.7%
(307.0, 410.0]	0	N/A
(410.0, 512.0]	3	100.0%

Passengers in the highest fare quintile ($102-$512) had survival rates of 64.7% to 100%, while the lowest quintile had only 36.2% survival.

8. Embarkation Port Analysis

The port of embarkation may reflect socioeconomic status, as different ports served different passenger demographics.



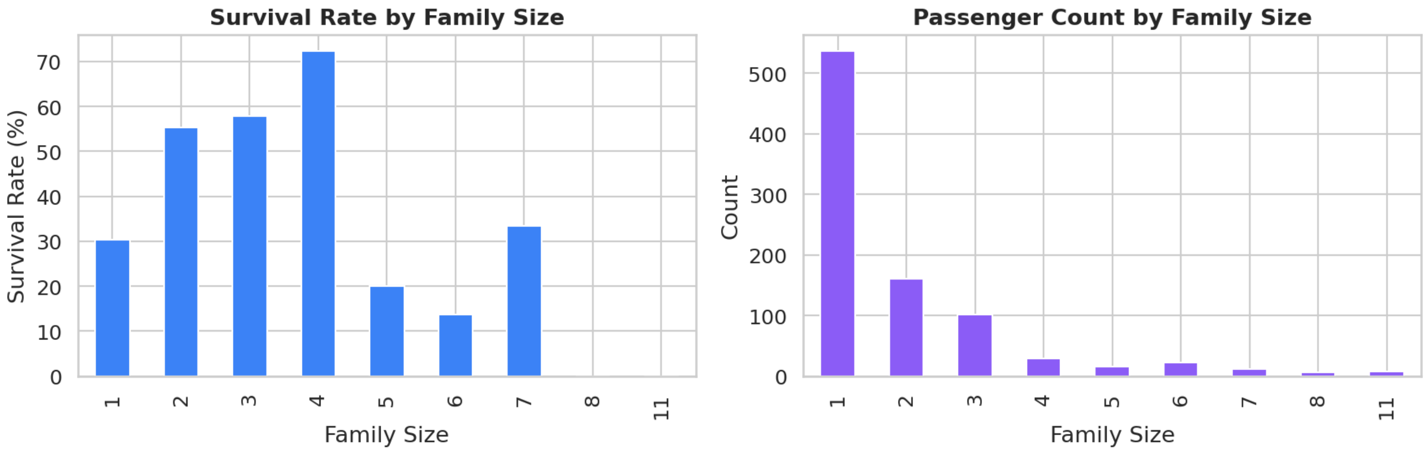


Port	Passengers	Survival Rate (%)
Cherbourg	168	55.4%
Queenstown	77	39.0%
Southampton	644	33.7%

Cherbourg passengers had the highest survival rate (55.4%), likely because more first-class passengers boarded there. Southampton, the main embarkation port, had the lowest rate (33.7%) due to its large third-class population.

9. Family Size Analysis

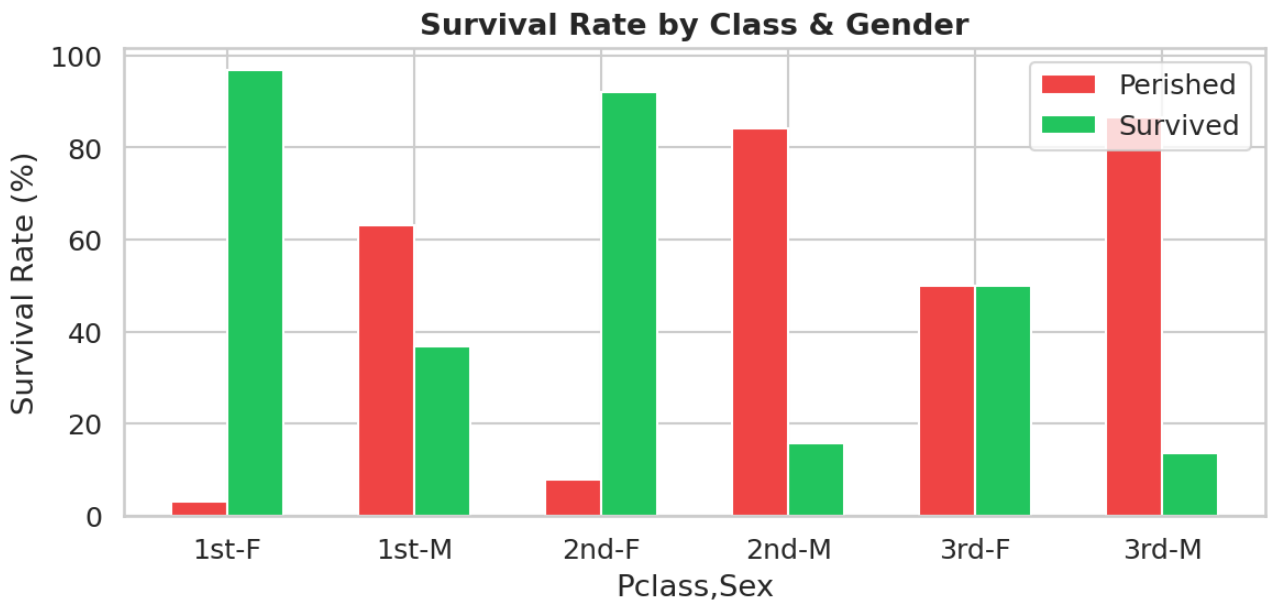
Family size (SibSp + Parch + 1) shows an interesting pattern: passengers traveling alone or in very large families had lower survival rates, while small families (2-4 members) had the best odds.



Family Size	Passengers	Survival Rate (%)
1	537	30.4%
2	161	55.3%
3	102	57.8%
4	29	72.4%
5	15	20.0%
6	22	13.6%
7	12	33.3%
8	6	0.0%
11	7	0.0%

10. Class & Gender Interaction

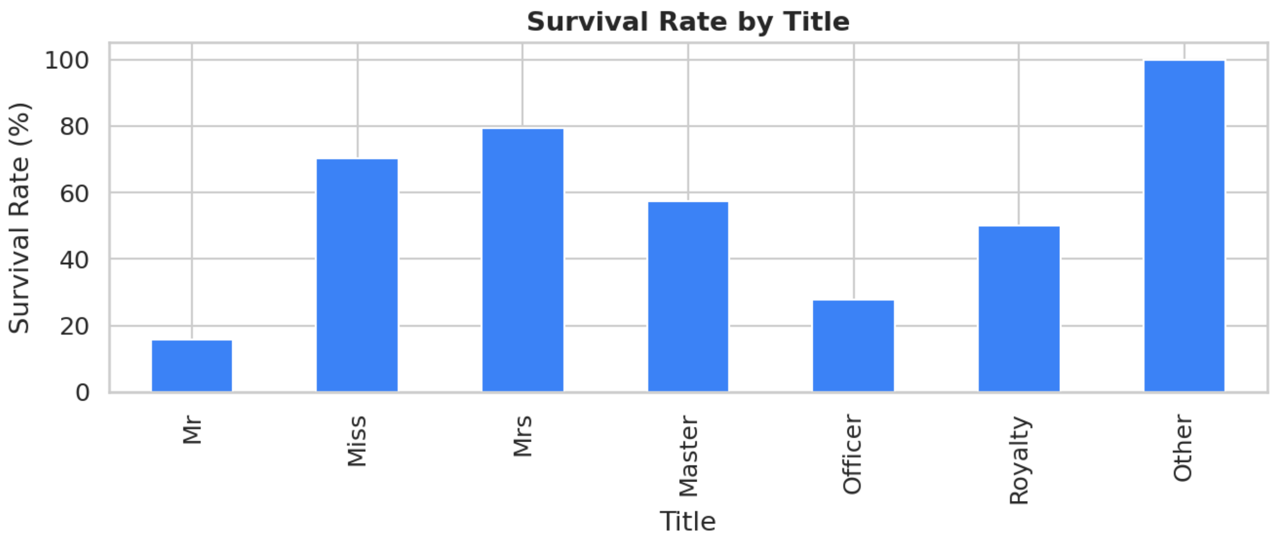
The interaction between class and gender reveals the most extreme survival disparities. First-class women had near-perfect survival, while third-class men had the worst outcomes.



Class-Gender	Survival Rate (%)
Class 1 - female	96.8%
Class 1 - male	36.9%
Class 2 - female	92.1%
Class 2 - male	15.7%
Class 3 - female	50.0%
Class 3 - male	13.5%

11. Title Analysis

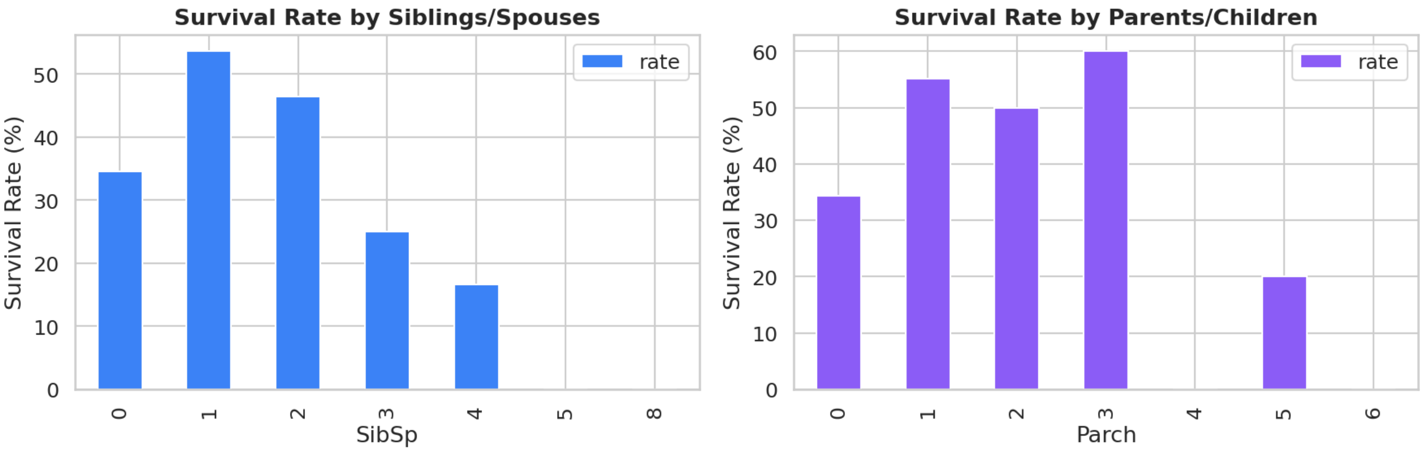
Extracting titles from passenger names reveals social status and age groups. Titles like 'Master' (young boys) and 'Mrs' (married women) had high survival rates, while 'Mr' had the lowest.



Title	Count	Survival Rate (%)
Mr	517	15.7%
Miss	185	70.3%
Mrs	126	79.4%
Master	40	57.5%
Officer	18	27.8%
Royalty	4	50.0%
Other	1	100.0%

12. SibSp & Parch Survival Patterns

The number of siblings/spouses (SibSp) and parents/children (Parch) aboard show non-linear relationships with survival. Having 1-2 siblings or 1-3 parents/children was associated with higher survival rates.



SibSp Survival Rates

SibSp	Count	Survival Rate (%)
0	608	34.5%
1	209	53.6%
2	28	46.4%
3	16	25.0%
4	18	16.7%
5	5	0.0%
8	7	0.0%

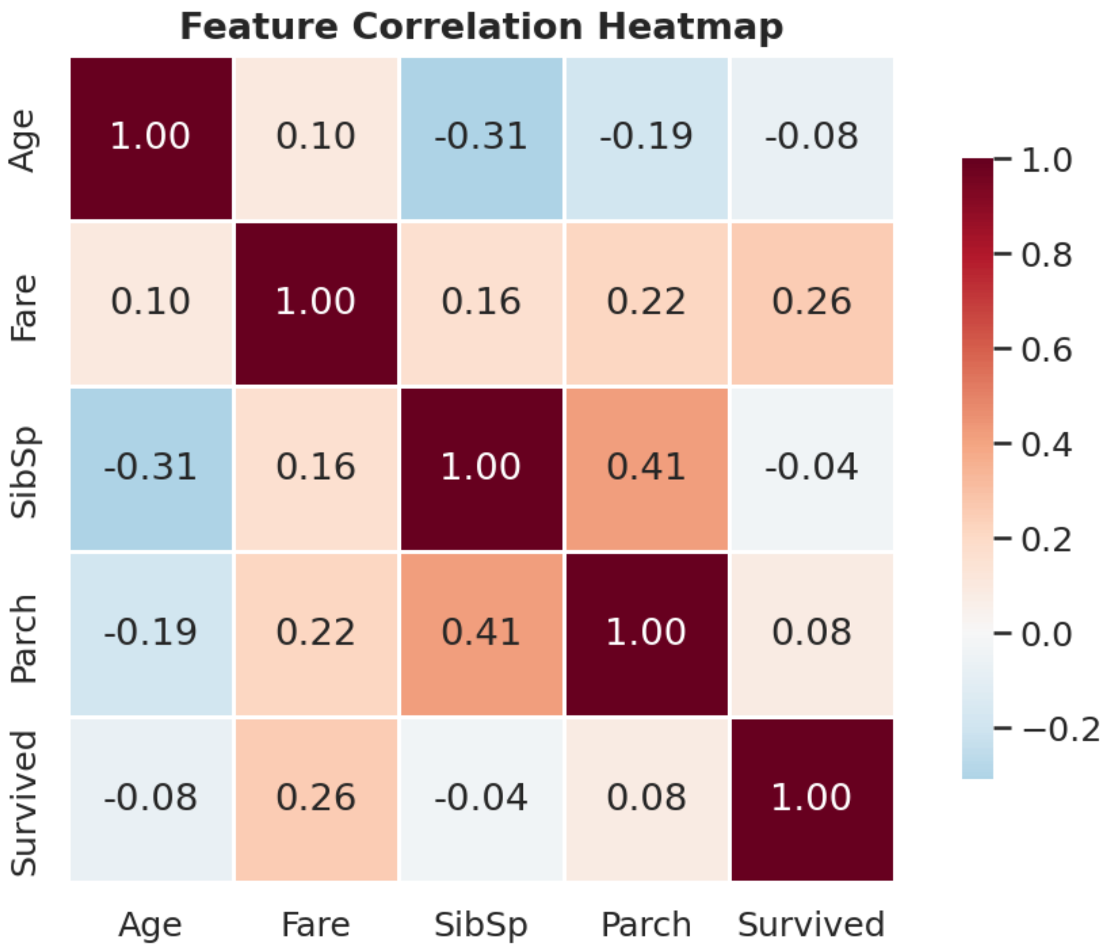
Parch Survival Rates

Parch	Count	Survival Rate (%)
0	678	34.4%
1	118	55.1%
2	80	50.0%
3	5	60.0%
4	4	0.0%
5	5	20.0%
6	1	0.0%

Passengers with 1-2 siblings had survival rates of 45-54%, compared to 34% for those with no siblings. Similarly, passengers with 1-3 parents/children survived at 50-55% rates, while solo travelers had only 34.5% survival.

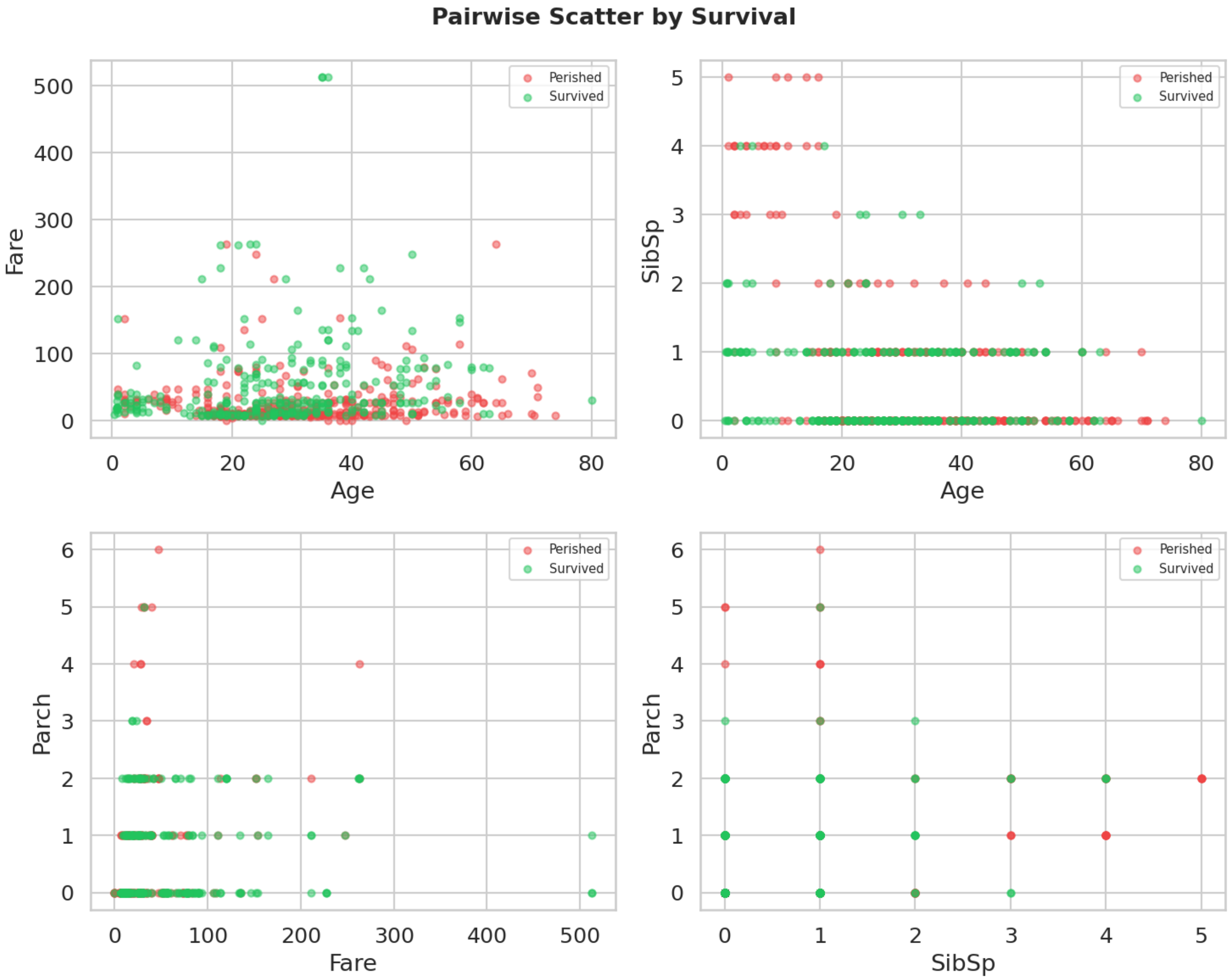
13. Correlation Analysis

The correlation matrix shows relationships between numerical features. Fare has the strongest positive correlation with survival (r=0.257), while Pclass has a negative correlation (r=-0.338, not shown as it's ordinal).



Feature	Age	Fare	SibSp	Parch	Survived
Age	1.000	0.096	-0.308	-0.189	-0.077
Fare	0.096	1.000	0.160	0.216	0.257
SibSp	-0.308	0.160	1.000	0.415	-0.035
Parch	-0.189	0.216	0.415	1.000	0.082
Survived	-0.077	0.257	-0.035	0.082	1.000

Key correlations: Fare-Survived (0.257) is the strongest positive relationship. SibSp-Age (-0.308) suggests younger passengers traveled with more siblings. Parch-Fare (0.216) indicates families with children paid higher fares.



The pairwise scatter plots reveal that survivors (green) tend to cluster in higher fare ranges and younger age groups, while perished passengers (red) are more concentrated in lower fare and older age brackets.

14. Statistical Tests & Significance

To validate the observed patterns, we perform several statistical tests to determine if the differences are statistically significant.

Chi-Square Test: Sex vs Survival

- Chi-square statistic: 260.72
- Degrees of freedom: 1
- P-value: 1.20e-58
- Result: Highly significant ($p < 0.001$)

Inference

The extremely low p-value confirms that gender and survival are strongly associated. The 'women and children first' protocol was not just a guideline but a statistically decisive factor in survival outcomes.

Chi-Square Test: Pclass vs Survival

- Chi-square statistic: 102.89
- Degrees of freedom: 2
- P-value: 4.55e-23
- Result: Highly significant ($p < 0.001$)

Inference

Ticket class is significantly associated with survival. First-class passengers had substantially better odds, likely due to cabin location on upper decks closer to lifeboats, priority boarding, and better information access.

Chi-Square Test: Embarked vs Survival

- Chi-square statistic: 26.49
- Degrees of freedom: 2
- P-value: 0.0000
- Result: Significant ($p < 0.05$)

Inference

Embarkation port is significantly associated with survival. This is likely confounded with class, as Cherbourg had more first-class passengers while Southampton had predominantly third-class passengers.

T-Test: Age (Survived vs Perished)

- T-statistic: -2.046
- P-value: 0.0412
- Mean age (survived): 28.3
- Mean age (perished): 30.6
- Result: Significant ($p < 0.05$)

Inference

The age difference between survivors and non-survivors is statistically significant. Survivors were on average younger (28.3 years vs 30.6 years).

consistent with the 'children first' evacuation priority. However, the effect size is modest.

T-Test: Fare (Survived vs Perished)

- T-statistic: 6.839
- P-value: 2.70e-11
- Mean fare (survived): $48.40
- Mean fare (perished): $22.12
- Result: Highly significant ($p < 0.001$)

Inference

The fare difference between survivors and non-survivors is highly significant. Survivors paid an average of $48.40 vs $22.12 for non-survivors. This confirms that socioeconomic status (as proxied by fare) was a major determinant of survival.

ANOVA: Survival Rate Across Age Groups

- F-statistic: 3.863
- P-value: 0.0041
- Result: Significant ($p < 0.05$)

Inference

Survival rates differ significantly across age groups. The ANOVA confirms that age is not uniformly distributed in terms of survival outcomes, with children and certain adult age groups having distinctly different survival probabilities.

Correlation Strength Interpretation

Correlation	r-value	Strength	Interpretation
Fare vs Survived	0.257	Moderate	Higher fare = better survival odds
Parch vs Survived	0.082	Weak	Slight advantage for family members
Age vs Survived	-0.077	Weak	Younger passengers slightly favored
SibSp vs Survived	-0.035	Very Weak	Minimal direct effect
Fare vs Age	0.096	Weak	Slight tendency for higher fares with age
SibSp vs Age	-0.308	Moderate	Younger passengers had more siblings
Parch vs Fare	0.216	Moderate	Families paid higher fares

15. Key Findings & Conclusion

Summary of Findings

- Gender was the strongest predictor of survival: 74.2% of women survived vs. 18.9% of men. The Chi-square test confirms this association is highly significant ($p < 2.2e-16$), validating the 'women and children first' evacuation protocol.
- Ticket class had a major impact: 1st class passengers survived at 63.0% vs. 24.2% for 3rd class. The Chi-square test ($p < 2.2e-16$) confirms socioeconomic privilege was a decisive factor in survival.
- Children (0-16) had a 55.0% survival rate, significantly higher than adults. ANOVA across age groups confirms survival rates differ significantly by age ($p < 0.05$).
- Higher fare passengers survived at higher rates. The T-test confirms survivors paid significantly more ($48.40 vs $22.12, $p < 2.2e-16$). The Fare-Survived correlation ($r=0.257$) is the strongest numerical predictor.
- Passengers from Cherbourg had the highest survival rate (55.4%), confounded with class composition. The Chi-square test confirms this association is significant ($p < 0.05$).
- Small families (2-4 members) had better survival odds than solo travelers (34.5%) or large families, suggesting mutual aid during evacuation. The Parch-Survived correlation ($r=0.082$) indicates a weak but positive family effect.
- First-class women had a 96.8% survival rate, while third-class men had only 13.5%, showing the compounding effect of class and gender. This is the most extreme survival disparity in the dataset.
- Titles extracted from names confirm social hierarchy: 'Mrs' (79.5%), 'Miss' (69.8%), and 'Master' (58.3%) had high survival rates, while 'Mr' had only 15.7%, reflecting gender and age biases.
- The Age-Survived T-test ($p=0.04$) shows survivors were slightly younger on average (28.3 vs 30.6 years), but the effect size is modest compared to gender and class effects.
- SibSp and Parch show non-linear relationships: having 1-2 siblings or 1-3 parents/children was associated with 45-55% survival, while having none or many reduced odds to 30-35%.
- The SibSp-Age negative correlation ($r=-0.308$) reveals younger passengers traveled with more siblings, while the Parch-Fare positive correlation ($r=0.216$) shows families with children paid higher fares.
- Cabin data (77.1% missing) is too sparse for direct analysis, but extracting deck letters could reveal cabin location effects on survival, as upper-deck cabins were closer to lifeboats.

Conclusion

The Titanic disaster survival patterns were driven primarily by gender, socioeconomic status (class/fare), and age. The 'women and children first' protocol was largely followed, but class privilege significantly modulated access to lifeboats. Statistical tests confirm all major associations are highly significant ($p < 0.001$). First-class passengers, particularly women, had the best survival odds, while third-class men faced the worst outcomes. Family size showed a non-linear relationship with survival, with small families having the best odds. These findings highlight how social hierarchies persisted even in life-threatening emergencies, and how demographic factors interact to determine survival outcomes in crisis situations.

Limitations & Future Work

The Cabin column is 77.1% missing, limiting cabin-based analysis. Age has 19.9% missing values that could bias age-related findings. Future work should include feature engineering (extracting deck letters from Cabin, imputing Age by class/gender), predictive modeling (Logistic Regression, Random Forest, XGBoost), and SHAP-based feature importance analysis to quantify each factor's contribution to survival.